

Chat Scratchy

An AI coding tutor + a live classroom dashboard

Kids learn to code with blocks while an AI tutor guides them — and teachers watch the whole class at a glance.

React 19 · Google Blockly · Express · PostgreSQL · SSE · LLM tutor

I Why Chat Scratchy?

The student's problem

- When a beginner gets stuck, **no help in the moment**
- Staring at an empty canvas, **unsure what's next**
- Frustration builds — and they **give up fast**

The teacher's problem

- One teacher has to watch **a whole class**
- Can't see what each child is **doing right now**
- No idea **who is stuck, or where**

Our answer

Give **every child** a patient tutor that actually **understands their code**, and give **every teacher** a dashboard showing the **whole class in real time**.

| Three pillars

1

AI tutor

A streaming chat tutor that **reads the student's blocks**, highlights them, gives tips, and **steps in automatically** when a child stalls.

2

Block coding

A Scratch-style Blockly workspace, a sprite stage and **live execution** — motion, looks, events, control and sensing blocks.

3

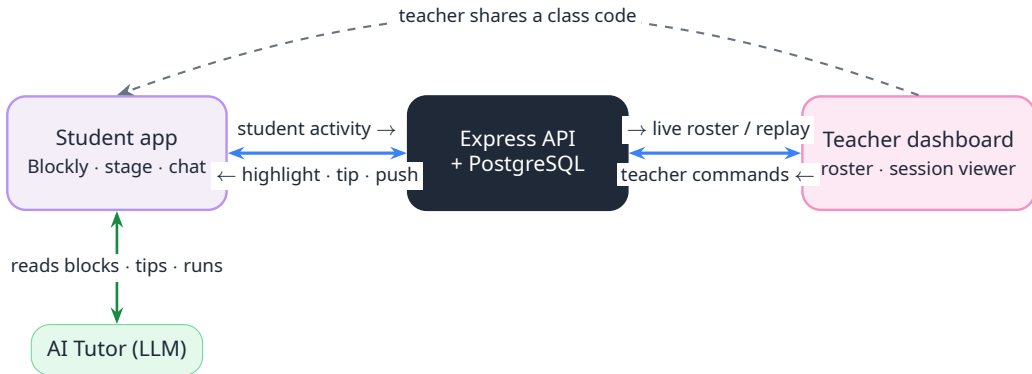
Teacher dashboard

Live class status, objective progress, chat replay — plus **two-way control** of any student's workspace.

How it works

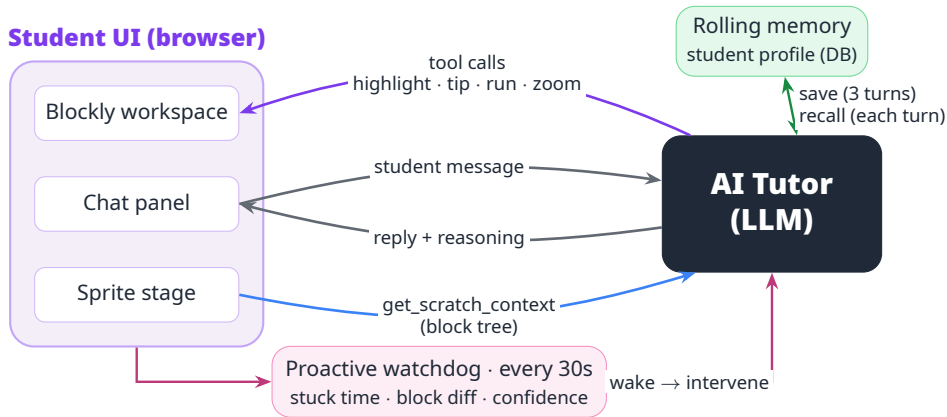
The moving parts, before we look at the screens

End to end, in one picture



A teacher shares a code; students join and code with the AI tutor; every action streams over **SSE** to the dashboard — and the teacher can stream hints and edits **back**.

Inside the AI tutor: agent $\Leftrightarrow$ UI



Each turn the tutor **reads the live block tree** and **acts on the UI** via tool calls; a **watchdog** wakes it when a student stalls.

The student experience

From joining a class to coding with an AI by your side

| One code, and you're in

- The teacher shares a **class code**
- Students enter the code and a name — **no signup, no password**
- Identity is remembered, so they **pick up where they left off**

A login form for 'Scratchy' with a pink-to-orange gradient background. The form is white with rounded corners. It includes a language toggle 'EN' in the top right, a welcome message '歡迎來到 Scratchy!', a prompt to enter the class code and name, two input fields for '課程代碼' and '你的名字', and a '開始學習' button at the bottom.

EN

歡迎來到 Scratchy!

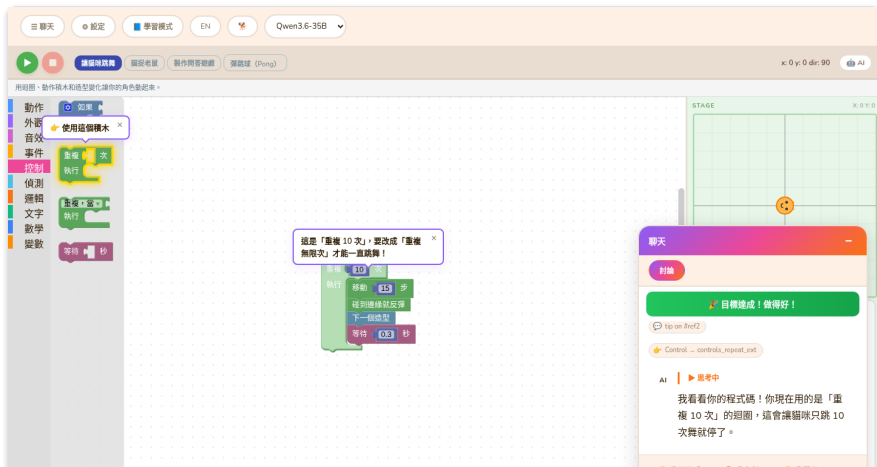
請輸入老師給你的課程代碼和你的名字。

課程代碼

你的名字

開始學習

The AI tutor reads your blocks



The tutor sees the “repeat 10” loop, **highlights the block**, leaves an **inline tip**, and congratulates the student when the objective is met.

Hit the green flag — the sprite comes alive

The screenshot shows the Scratch 3.0 web interface. On the left is the 'Spritel' (Sprites) palette with categories like '動作' (Actions), '外觀' (Appearance), '音效' (Sounds), '事件' (Events), '控制' (Control), '偵測' (Sensing), '邏輯' (Logic), '文字' (Text), '數學' (Math), and '變數' (Variables). The main workspace shows a script for a sprite named '彈珠球 (Pong)'. The script starts with a '當綠旗被點擊時' (When green flag is clicked) event, followed by a '重複 10 次' (Repeat 10 times) loop. Inside the loop, the steps are: '移動 15 步' (Move 15 steps), '碰到邊緣就反彈' (Bounce when hit edge), '下一個造型' (Next costume), and '等待 0.3 秒' (Wait 0.3 seconds). A tooltip points to the '重複 10 次' block with the text: '這是「重複 10 次」，要改成「重複無限次」才能一直跳舞！' (This is 'Repeat 10 times', to make it dance continuously, it should be changed to 'Repeat infinitely'!). On the right, a chat window titled '聊天' (Chat) shows a message: '目標達成！做得好！' (Goal achieved! Good job!). Below the chat, there are buttons for '我很困惑' (I'm confused), '我在想' (I'm thinking), and '我懂了' (I got it). At the bottom, there is a '傳送訊息...' (Send message...) input field.

Blocks **compile to code and run instantly**

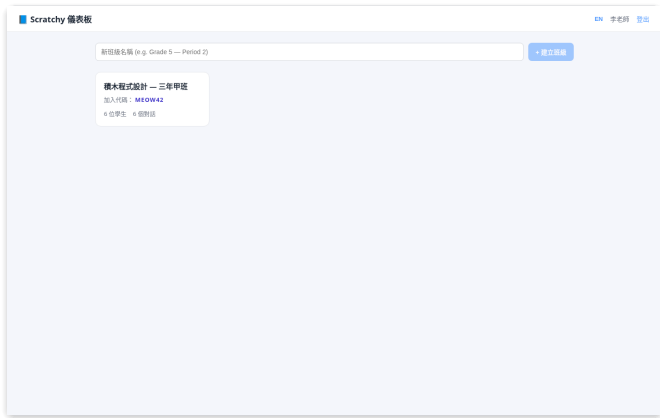
- The sprite moves, talks and changes costume on the **stage**
- Green-flag, key, click and broadcast **events**
- Multiple built-in **costumes and sprites**

The teacher dashboard

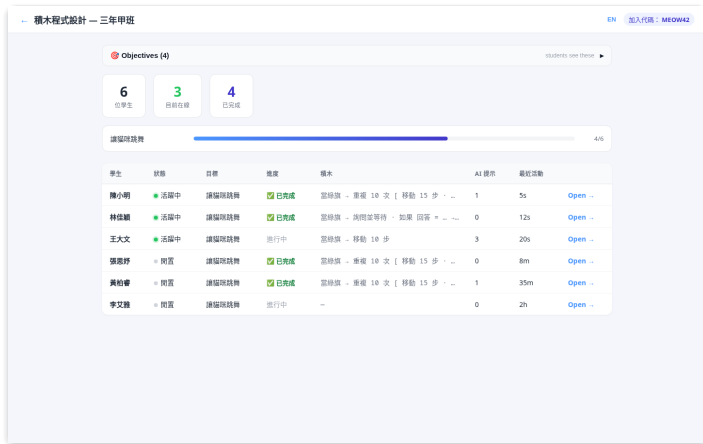
The whole class's progress, on one screen

Class management

- Secure login (httpOnly JWT)
- **Create a class** and generate a join code in one click
- Live **student and session counts** per class



The class, live



- Who's online (active in 90s)

Per-objective **completion progress**

AI-intervention count & latest blocks

Drill into one student

陳小明 · 積木程式設計 — 三年甲班

EN 1 AI hints 100% effective 0bcb697c-c08b-4e71-995f-1274c841b15a

AI memory 小明學習動作積木很快，充滿自信。他喜歡製作動畫，會先自己嘗試再求助。可鼓勵他在堆疊積木前先規劃迴圈，並養成加上註解的習慣。 updated 6/9/2026, 3:45:17 PM

CHAT TRANSCRIPT

You 怎麼讓貓咪一直動？

AI 好問題！🐱 用「重複」迴圈最適合。試著從「控制」抽屜拿一個重複 10 次的積木，把你的移動 10 步放進去。你覺得點綠旗後會發生什麼事呢？

You 它會動可是跑出畫面了

AI 很棒的觀察！要讓它留在舞台上，在移動後面加一個碰到邊緣就反彈積木，貓咪撞到牆就會轉向。要不要再加一個下一個造型，讓牠看起來像在跳舞？

You 哇 成功了！！牠在跳舞

AI 太棒了，小明！👏 你把迴圈、動作和造型變化組合在一起，這就是一個真正的動畫。下一個挑戰：可以讓牠每次移動之間都「等待」一下下，讓舞步更順嗎？

Blocks Code Timeline

Highlight Tip Clear Edit mode

```
when flag clicked
repeat (10) times
do
  move (15) steps
  if on edge, bounce
  next costume
  wait (0.3) seconds
```

Chat replay on the left, the **live workspace** on the right; the *AI memory* banner is the rolling profile the tutor builds across sessions.

Two-way live control

The teacher can **step into** a student's workspace in real time:

- **Highlight** a block
- Leave an **inline tip** on a block
- **Push** an edited workspace straight to the student

And inspect the JavaScript Blockly generates.

陳小明·積木程式設計 — 三年甲班

AI memory 小明學習動作積木很快，充滿自信。他喜歡製作動畫，會先自己嘗試再求助。可鼓勵他在堆疊積木前先規劃迴圈，並養成加上註解的習慣。

CHAT TRANSCRIPT

You 怎麼讓貓咪一直動？

AI 好問題！🐾 用「重複」迴圈最適合。試著從「控制」抽屜拿一個 **重複 10 次** 的積木，把你的 **移動 10 步** 放進去。你覺得點綠旗後會發生什麼事呢？

You 它會動可是跑出畫面了

AI 很棒的觀察！要讓它留在舞台上，在移動後面加一個 **碰到邊緣就反彈** 積木，貓咪碰到邊就會轉向。要不要再加一個 **下一個造型**，讓她看起來像在跳舞？

You 哇 成功了！！牠在跳舞

AI 太棒了，小明！🐾 你把迴圈、動作和造型變化組合在一起，這就是一個真正的動畫。下一個挑戰：可以讓牠每次移動之間都「等待」一下下，讓舞步更順嘴？

Blocks Code Timeline

```
// 當綠旗被點擊  
for (let i = 0; i < 10; i++) {  
  await window.__scratchMove(15);  
  await window.__scratchBounce();  
  window.__scratchNextCostume();  
  await window.__scratchWait(0.3);  
}
```

Activity timeline & intervention impact

← 陳小明·積木程式設計 — 三年甲班

AI memory 小明學習動作積木很快，充滿自信。他喜歡製作動畫，會先自己嘗試再求助。可鼓勵他在堆疊積木前先規劃圖面，並養成加上註解的習慣。 updated 6/9/2026, 3:45:17 PM

CHAT TRANSCRIPT

You 怎麼讓貓咪一直動？

AI 好問題！🐾 用「重複」迴圈最適合。試著從「控制」抽屜拿一個 重複 10 次 的積木，把你的 移動 10 步 放進去。你覺得點綠旗後會發生什麼事呢？

You 它會動可是跑出畫面了

AI 很棒的觀察！要讓它留在舞台上，在移動後面加一個 碰到邊緣就反彈 積木，貓咪撞到牆就會轉向。要不要再加一個 下一個造型，讓牠看起來像在跳舞？

You 哇 成功了！！牠在跳舞

AI 太棒了，小明！🐾 你把迴圈、動作和造型變化組合在一起，這就是一個真正的動畫。下一個挑戰：可以讓牠每次移動之間都「等待」一下下，讓舞步更順嗎？

Blocks Code Timeline

需要幫忙嗎？試著執行程式看看會發生什麼事。 helped

pattern: proactive

student_message

run

block_change

confidence (thinking)

run

objective_complete

confidence (good)

- **Messages, runs, stuck states, confidence and completions** in time order
- **Every proactive AI intervention** and its outcome
- Auto-computed **effectiveness** (helped / resolved)

I Architecture

Frontend

- React 19 + TypeScript + Vite
- Google Blockly v12 (custom Scratch blocks + JS generator)
- Streaming chat with reasoning display

Backend

- Express.js + cookie JWT auth
- PostgreSQL via Prisma ORM

Realtime

- Server-Sent Events, both directions
- Teacher $\Leftrightarrow$ student command channel

AI tutor

- OpenAI-compatible LLM backend
- Tool calls: read blocks, highlight, tip, run
- Background watchdog that **intervenes when stuck**
- Rolling student memory across sessions

Recap

- Every child gets an AI tutor that **understands their blocks**
- A complete **block-coding** environment with a live stage
- Teachers get a **real-time, whole-class** dashboard with two-way control
- The AI builds a **rolling profile** of each student over time

Kids code with blocks, an AI guides them, and teachers see everything.